

A Literature-Implied Negative Answer to the Boundary-Lipschitz Harmonic-Extension Question

Math discovery packet for arXiv:1405.0617

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1 Status and source

Status: Literature-implied negative answer with an explicit disk witness. This packet records a known endpoint-regularity implication, not a new original counterexample.

Kolesnikov and Milman ask in Remark 13 on PDF page 11 of arXiv:1405.0617v2 whether it is enough to control the variance of harmonic functions whose restrictions to the boundary are 1-Lipschitz. They explain that their intended use of the weak L^2 - L^∞ Poincare inequality would require control of the Lipschitz constant throughout the convex domain, and identify the uncontrolled normal derivative as the obstacle [1].

The boundary-to-interior Lipschitz property is false already for the unit disk. The calculation below also gives a smooth family showing that no uniform estimate can hold even when every harmonic function is smooth on a neighborhood of the closed disk.

2 Literature identification

The endpoint phenomenon is classical. Hile and Stanoyevitch prove that, for a bounded C^2 domain and Lipschitz Dirichlet data, the harmonic extension has a pointwise gradient bound of logarithmic order in the inverse distance to the boundary [2, Theorem 2]. Katzourakis later derives global $W^{1,p}$ -bounds for every finite p , explicitly from this logarithmic estimate [3].

For the unit disk, Olofsson's Theorem 6.7 and Proposition 6.8 give the endpoint Lipschitz characterization in terms of the conjugate-function (Hilbert transform) condition on the boundary derivative [4]. The example below is the elementary Fourier witness: its boundary derivative is a sign function and the associated normal derivative has logarithmic blow-up.

These supporting sources do not claim to answer the source remark. The relation is therefore filed as an implied literature answer, not as an explicit literature answer.

3 Explicit unit-disk witness

Proposition 1. *There is a real harmonic function $H \in C(\overline{\mathbb{D}}) \cap C^\infty(\mathbb{D})$ whose boundary restriction is 1-Lipschitz for the ambient Euclidean metric on $\mathbb{T}$, but H is not Lipschitz on $\mathbb{D}$.*

Moreover, there are functions H_ρ , $0 < \rho < 1$, harmonic on neighborhoods of $\mathbb{D}$, such that

$$\text{Lip}(H_\rho|_{\mathbb{T}}) \leq 1 \quad \text{and} \quad \text{Lip}(H_\rho|_{\overline{\mathbb{D}}}) \longrightarrow \infty \quad (\rho \uparrow 1).$$

Proof. For $-\pi \leq t \leq \pi$, put

$$g(e^{it}) = \frac{2}{\pi}|t|.$$

This is the scaled geodesic distance on $\mathbb{T}$ from 1. If the shorter angular distance between e^{it} and e^{is} is $\delta \in [0, \pi]$, then

$$|g(e^{it}) - g(e^{is})| \leq \frac{2}{\pi} \delta \leq 2 \sin(\delta/2) = |e^{it} - e^{is}|,$$

where the middle inequality follows from concavity of $\sin$ on $[0, \pi/2]$. Thus $\text{Lip}(g) \leq 1$.

The Fourier series of g is

$$g(e^{it}) = 1 - \frac{8}{\pi^2} \sum_{m=0}^{\infty} \frac{\cos((2m+1)t)}{(2m+1)^2}.$$

Consequently its Poisson extension is

$$H(re^{it}) = 1 - \frac{8}{\pi^2} \sum_{m=0}^{\infty} \frac{r^{2m+1} \cos((2m+1)t)}{(2m+1)^2}.$$

The series is harmonic and smooth on compact subsets of $\mathbb{D}$, and the absolute convergence of the boundary Fourier series gives $H \in C(\overline{\mathbb{D}})$ with boundary value g . Along the positive real radius, termwise differentiation for $0 < r < 1$ gives

$$\begin{aligned} \partial_r H(r) &= -\frac{8}{\pi^2} \sum_{m=0}^{\infty} \frac{r^{2m}}{2m+1} \\ &= -\frac{4}{\pi^2 r} \log\left(\frac{1+r}{1-r}\right), \end{aligned}$$

because $\sum_{m \geq 0} r^{2m+1}/(2m+1) = \text{arctanh } r$. The radial derivative tends to $-\infty$ as $r \uparrow 1$, so H cannot be Lipschitz on $\mathbb{D}$.

For the smooth version, set $H_\rho(z) = H(\rho z)$. This is harmonic in the disk of radius $1/\rho > 1$. Its boundary trace is the Poisson average $P_\rho * g$. Positivity and rotational invariance of the Poisson kernel imply

$$|(P_\rho * g)(e^{it}) - (P_\rho * g)(e^{is})| \leq \int_{\mathbb{T}} P_\rho(du) |g(e^{i(t-u)}) - g(e^{i(s-u)})| \leq |e^{it} - e^{is}|.$$

Hence $\text{Lip}(H_\rho|_{\mathbb{T}}) \leq 1$. On the other hand,

$$|\partial_r H_\rho(1)| = \frac{4}{\pi^2} \log\left(\frac{1+\rho}{1-\rho}\right) \rightarrow \infty.$$

The Lipschitz seminorm on $\overline{\mathbb{D}}$ dominates this inward directional derivative, proving the second assertion. $\square$

4 Scope and verification

This closes only the proposed interior-Lipschitz route in Remark 13. It does not settle the KLS conjecture and does not preclude a direct variance estimate for the boundary-Lipschitz harmonic subclass.

The bounded novelty check searched the run indexes for the arXiv id, title, exact topic, and core terms; it also searched the web for the exact source phrase, the source id/title, and harmonic-extension endpoint variants. The search found the classical logarithmic estimate [2], the finite- p

consequence [3], and the endpoint unit-disk characterization [4], but no paper explicitly advertising a resolution of Kolesnikov–Milman Remark 13.

The accompanying script `code/verify_disk_formula.py` compares the Fourier partial sums with the closed derivative formula at several radii and checks the predicted monotone blow-up. This is a sanity check only; the proof above is exact.

Human-review recommendation. Check the intended quantifier in the phrase “bounded Lipschitz constant.” The single limit function refutes global Lipschitz regularity for continuous Lipschitz boundary data, while the smooth family refutes every uniform boundary-to-interior estimate in the smooth class.

References

- [1] A. V. Kolesnikov and E. Milman, *Remarks on the KLS conjecture and Hardy-type inequalities*, GAFA Seminar Notes, arXiv:[1405.0617](#).
- [2] G. N. Hile and A. Stanoyevitch, *Gradient bounds for harmonic functions Lipschitz on the boundary*, *Applicable Analysis* 73 (1999), 101–113, doi:[10.1080/00036819908840767](#).
- [3] N. Katzourakis, *A remark on global $W^{1,p}$ bounds for harmonic functions with Lipschitz boundary values*, *Glasnik Matematicki* 52 (2017), 133–141, arXiv:[1601.00190](#).
- [4] A. Olofsson, *Lipschitz continuity for weighted harmonic functions in the unit disc*, *Complex Variables and Elliptic Equations* 65 (2020), 1630–1660, doi:[10.1080/17476933.2019.1669572](#).